UNIVERSITY OF BIRMINGHAM | GRAVITATIONAL WAVE ASTRONOMY

OBSERVATIONAL COSMOLOGY

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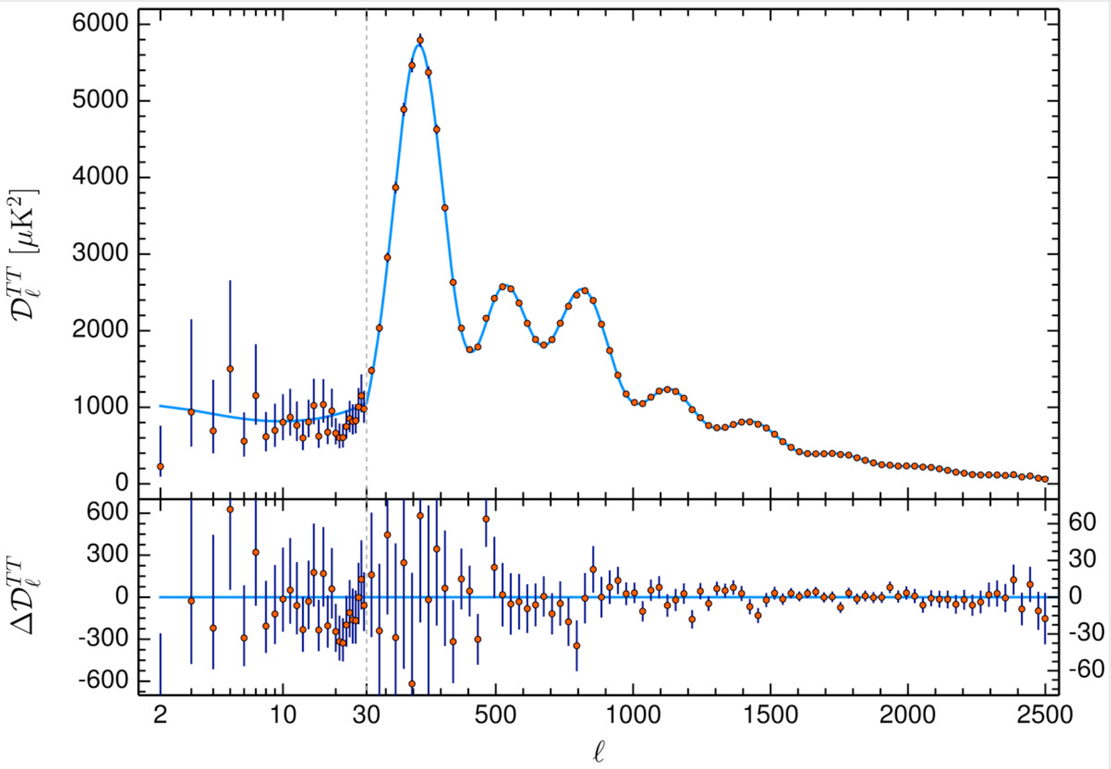
RECAP & SUMMARY: CMB ANISOTROPIES

- Fundamentally, the observed temperature fluctuations in the CMB originated from **density perturbations** in the early universe, i.e. density inhomogeneity is converted into temperature anisotropy
- We analyse the fluctuations statistically by computing their **angular power spectrum** (two-point correlation function)
- **Different physical effects** source temperature anisotropies on different scales:
  - Doppler shifting due to the bulk fluid motion (on all scales)
  - **Sachs-Wolfe effect** (gravitational redshift) dominates on scales larger than the sound horizon distance
  - **BAOs** dominate on scales smaller than the sound horizon distance at LS



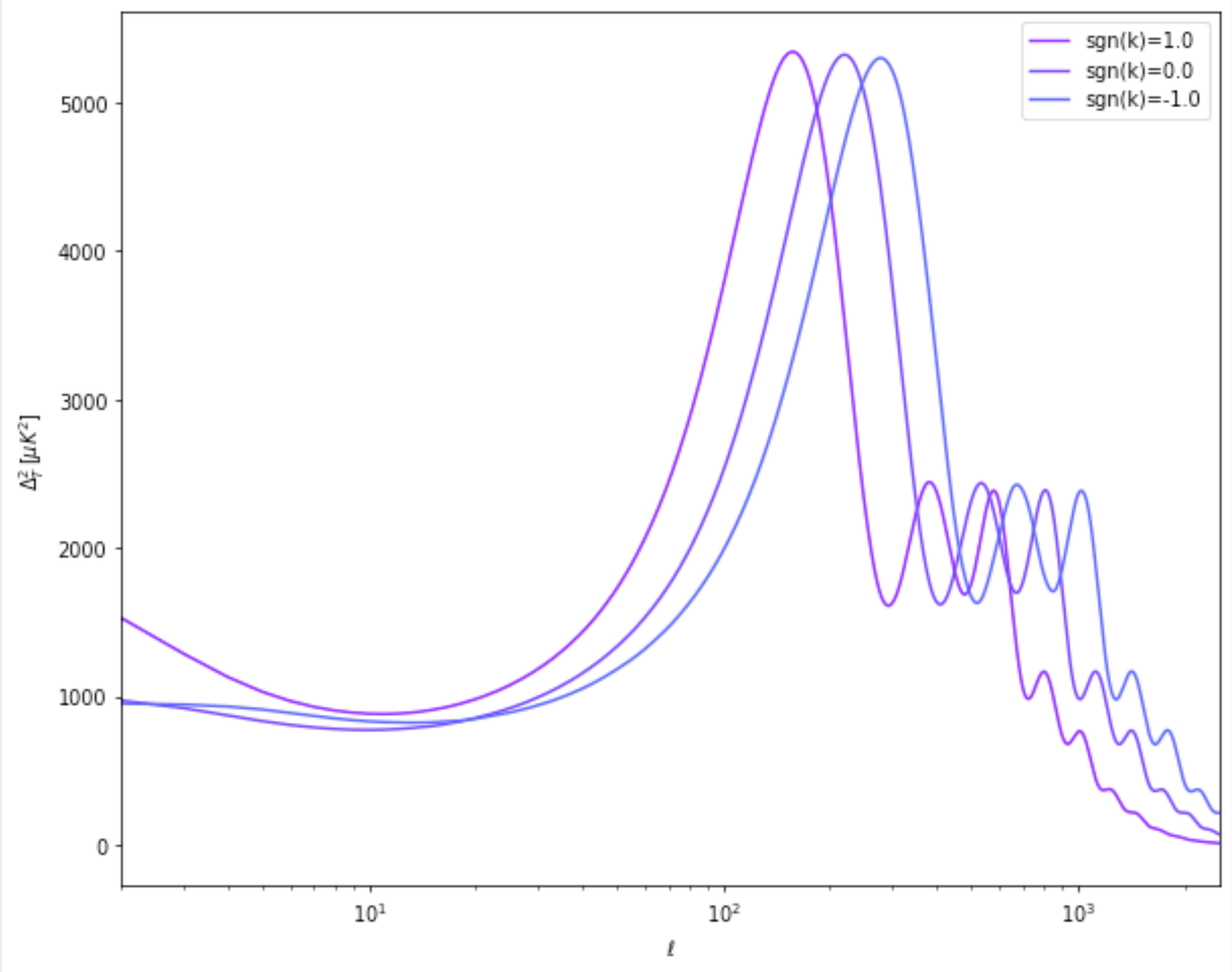
POWER SPECTRUM & COSMOLOGICAL PARAMETERS

- How does the power spectrum change if we change the physics in the early universe, which in turn would lead to different density parameters or spatial curvature?

- Measuring the power spectrum with high precision and comparing it to theoretical predictions allows us to constrain (some) cosmological parameters to 1% accuracy.

- Consider base ΛCDM and introduce some spatial curvature:
- A closed universe shifts the acoustic peak to larger angular scale (lower multipole)

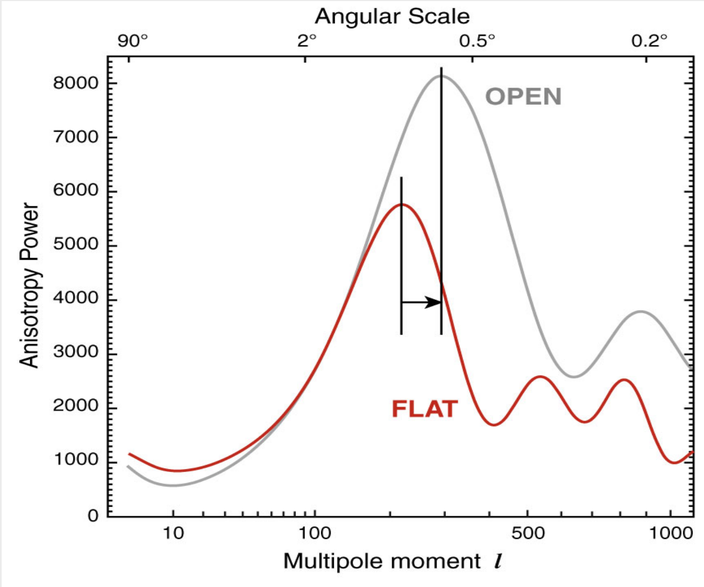
- Note: I fixed H_0 and set $\Omega_\Lambda = 0$
- Code: CAMB https://camb.readthedocs.io/en/latest/

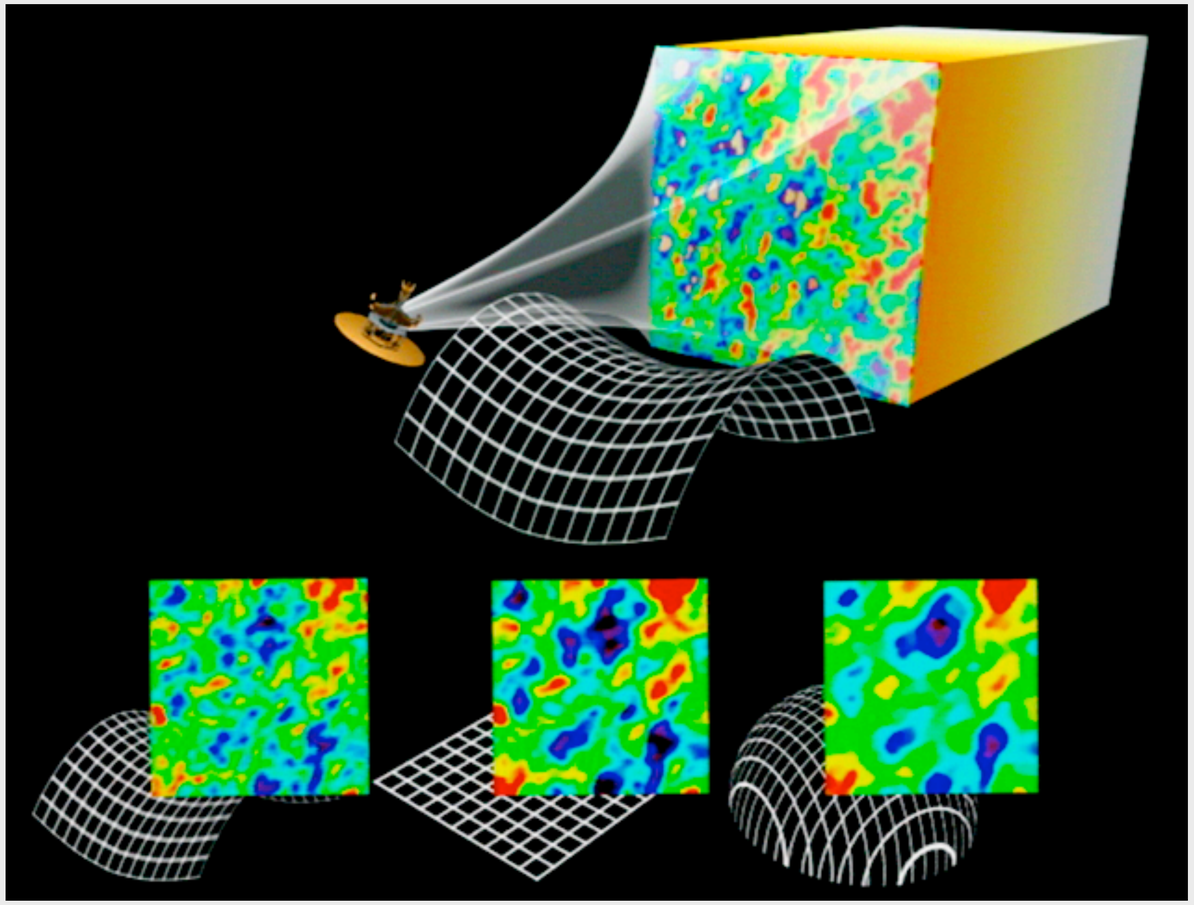


POWER SPECTRUM & COSMOLOGICAL PARAMETERS

Changing the spatial curvature changes the path of light propagation:

- For a spherical geometry, this results in larger hot/cold spots, i.e. the first peak moves to lower angular scale
- For a hyperbolic geometric, this results in smaller hot/cold spots, i.e. the first peak moves to smaller angular scale (higher multipole)

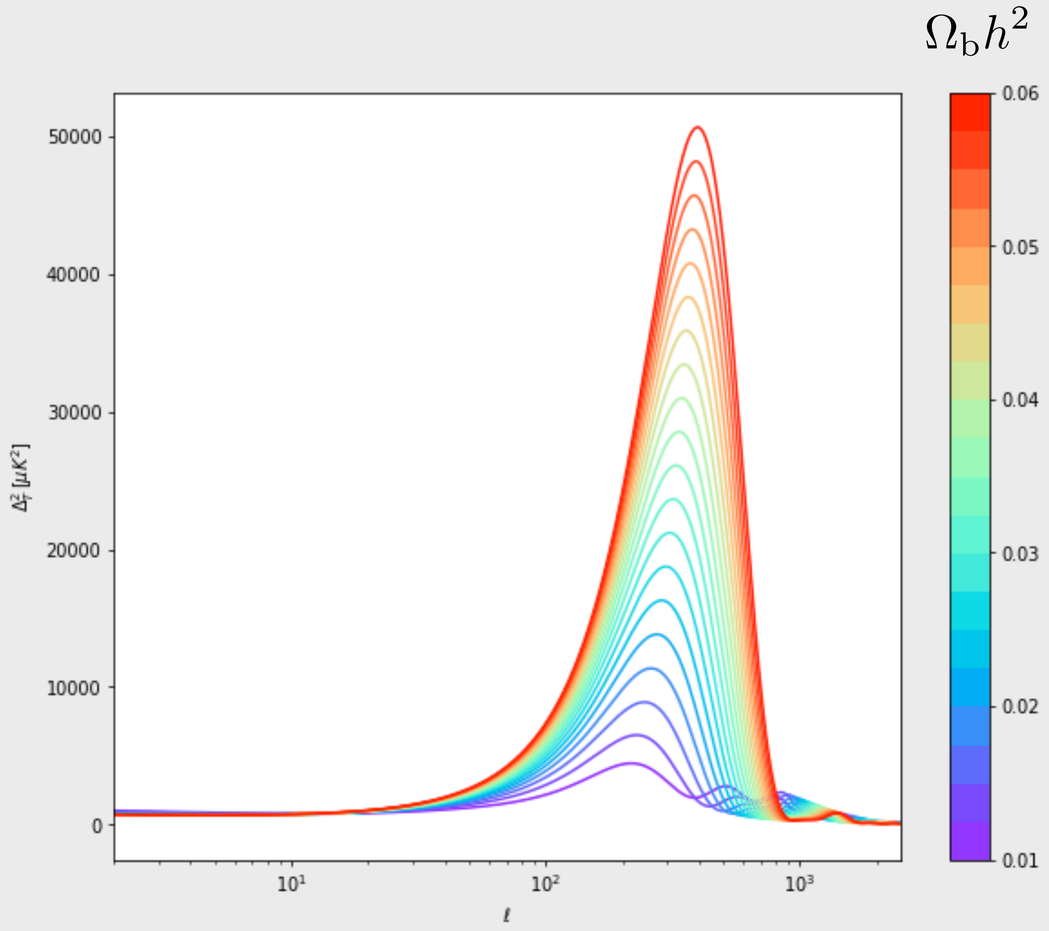


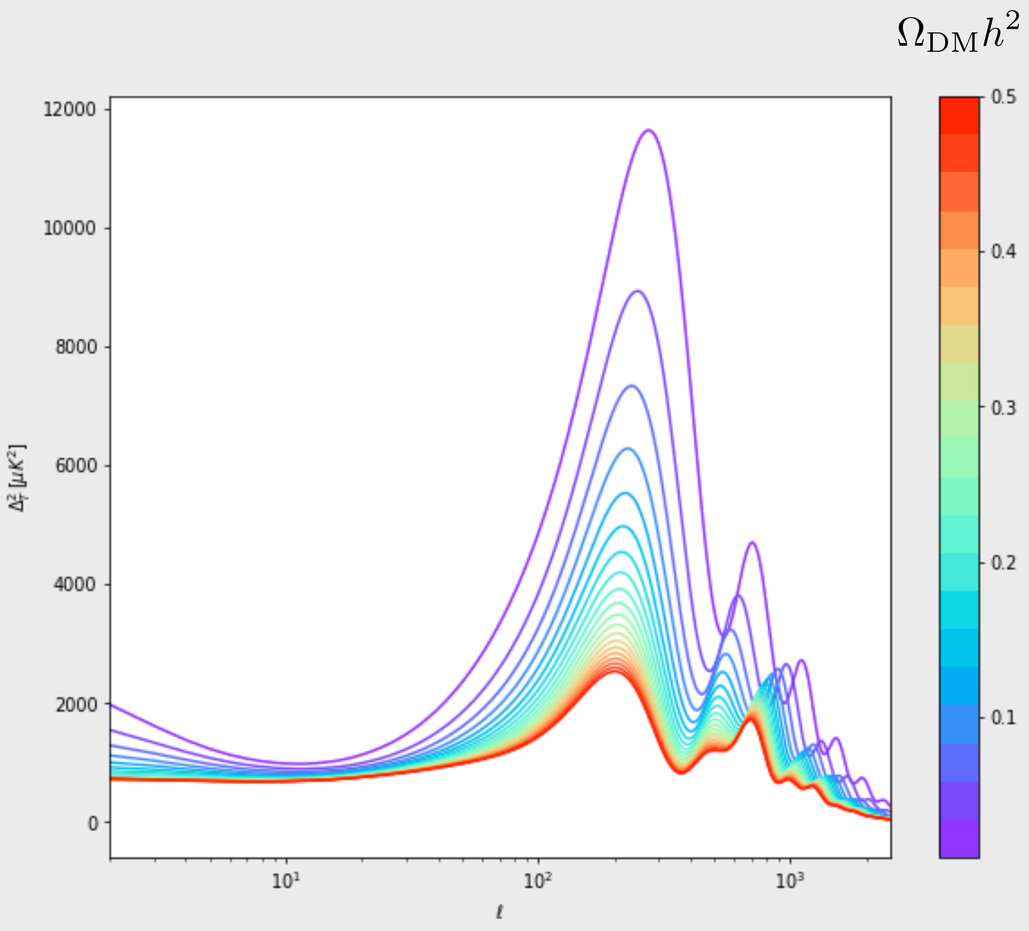


[Credit: WMAP]

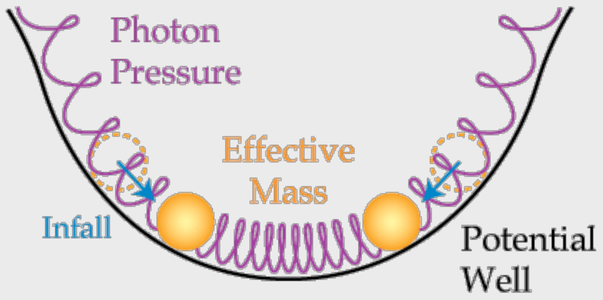
POWER SPECTRUM & COSMOLOGICAL PARAMETERS

- Assuming a spatially flat universe:





POWER SPECTRUM & COSMOLOGICAL PARAMETERS

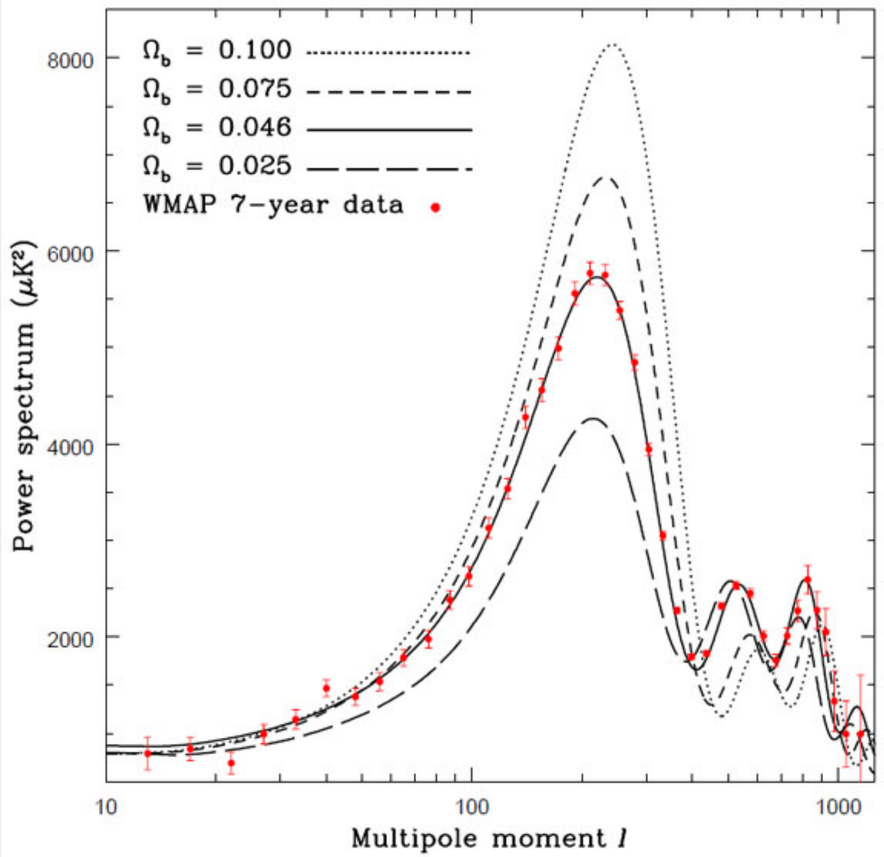


- Changing the **matter density (baryonic + DM)** impacts
  - The height of the first peak
  - The number of peaks

- <u>Increasing the baryon density</u> leads to a more massive effective "spring", hence increasing the power in the first peak and suppressing higher multipoles (similar to a spring in a gravitational field); the peak also shifts to the right as the effective mass of the spring is increased which changes the wavelength

- <u>Increasing the DM density</u> leads to a decrease in the power in the first peak due to a stronger gravitational potential against which the baryons are working

- BUT: we cannot directly measure H_0 from the CMB



MEASURING H0 FROM THE CMB: SUMMARY

- Characteristic scale at LS: (proper) sound horizon distance
  - Comoving sound horizon distance:

$$r_s(t_{\rm LS}) = \frac{d_s(t_{\rm lS})}{a(t_{\rm LS})} = (1+z_{\rm LS})d_s(t_{\rm LS}) = \int_{z_{\rm LS}}^{\infty} \frac{c_s(z)}{H(z)}dz$$

  - Sound speed in photon-baryon plasma depends on η.
- For $z \gg 1$:

$$H(z) \approx 100\sqrt{\Omega_{m,0}h^2(1+z)^3 + \Omega_{r,0}h^2(1+z)^4}$$

Determined by fitting ΛCDM to the CMB power spectrum.

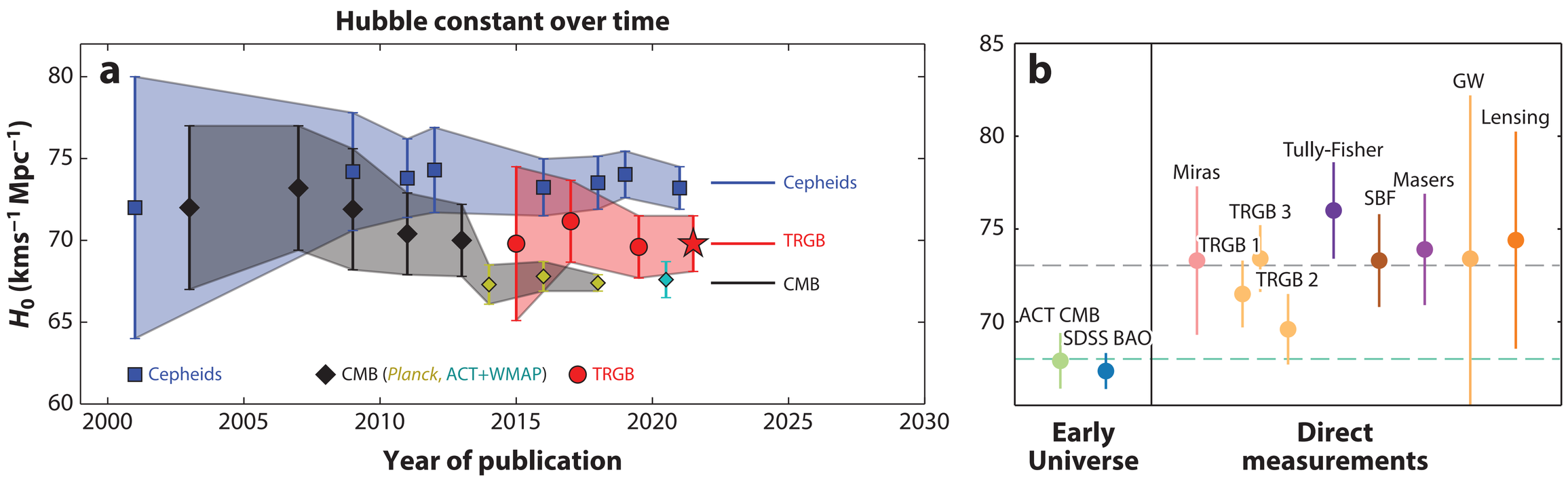
- Using Planck values: $r_s \approx 145\,\mathrm{Mpc}$
- Angular size precisely measured from first acoustic peak ($\ell \approx 220$):

$$\theta_s = \frac{d_s(t_{\rm LS})}{d_A(t_{\rm LS})} \equiv \left.\frac{r_s}{d_A^{\rm com}}\right|_{z_{\rm LS}} \approx 0.7^\circ$$

The comoving angular diameter distance depends on H_0!

THE HUBBLE TENSION

- Recent review: https://www.annualreviews.org/content/journals/10.1146/annurev-astro-052622-033813



HUBBLE TENSION: POSSIBLE SOLUTIONS

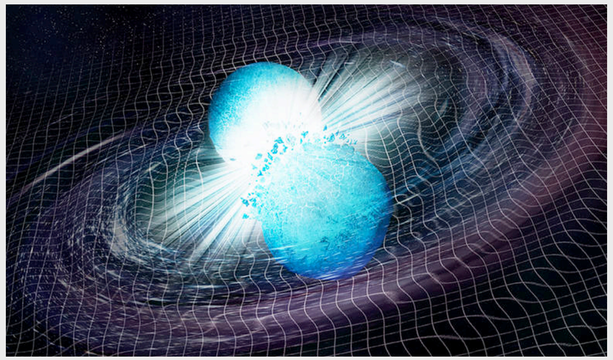
- Systematic errors:
  - In the CMB measurements (early time)
  - In the late-time probes (distance ladder)
  - Both unlikely to be the sole solutions as independent measurements are affected by different systematics

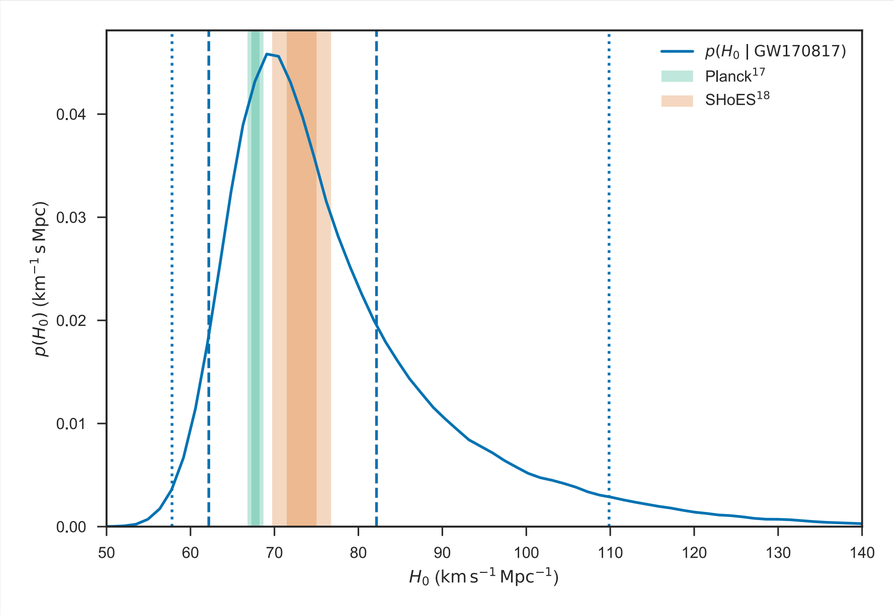
- New physics beyond the standard model of cosmology, i.e. ΛCDM
  - ATTENTION: To reconcile the early time and late time measurements, new physics can only affect the decade of expansion before recombination
  - Some options that do not worsen the fit to the current data:
    - Scalar field as an early DE component
    - Extending the radiation sector in the early universe, e.g. neutrino self-interaction and/or additional neutrino species

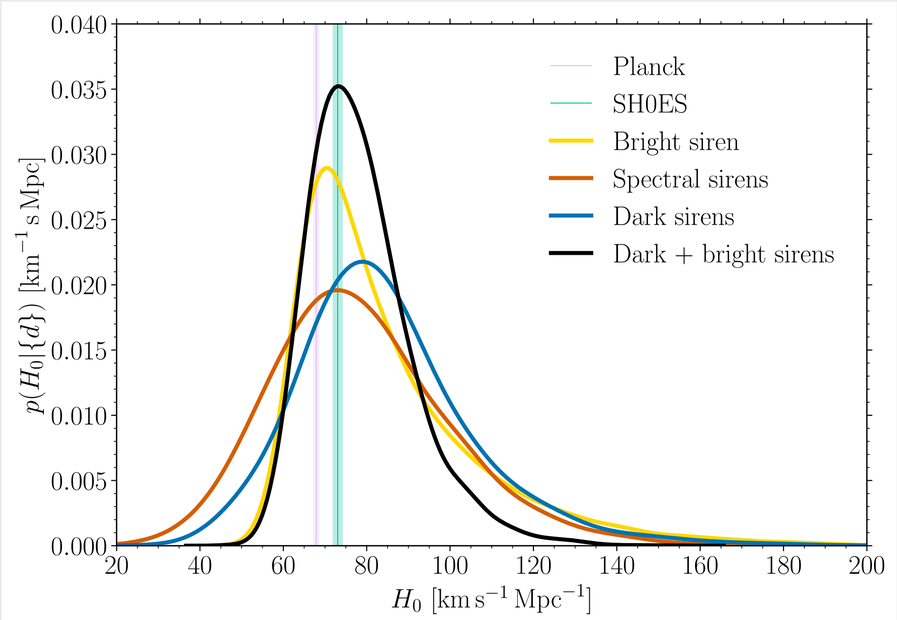
New physics must leave verifiable imprints in the data that can be probed with future observations!

INDEPENDENT PROBE: GRAVITATIONAL WAVES

- We can independently measure the luminosity distance from the observation of gravitational waves from colliding neutron stars (and black holes but different method)



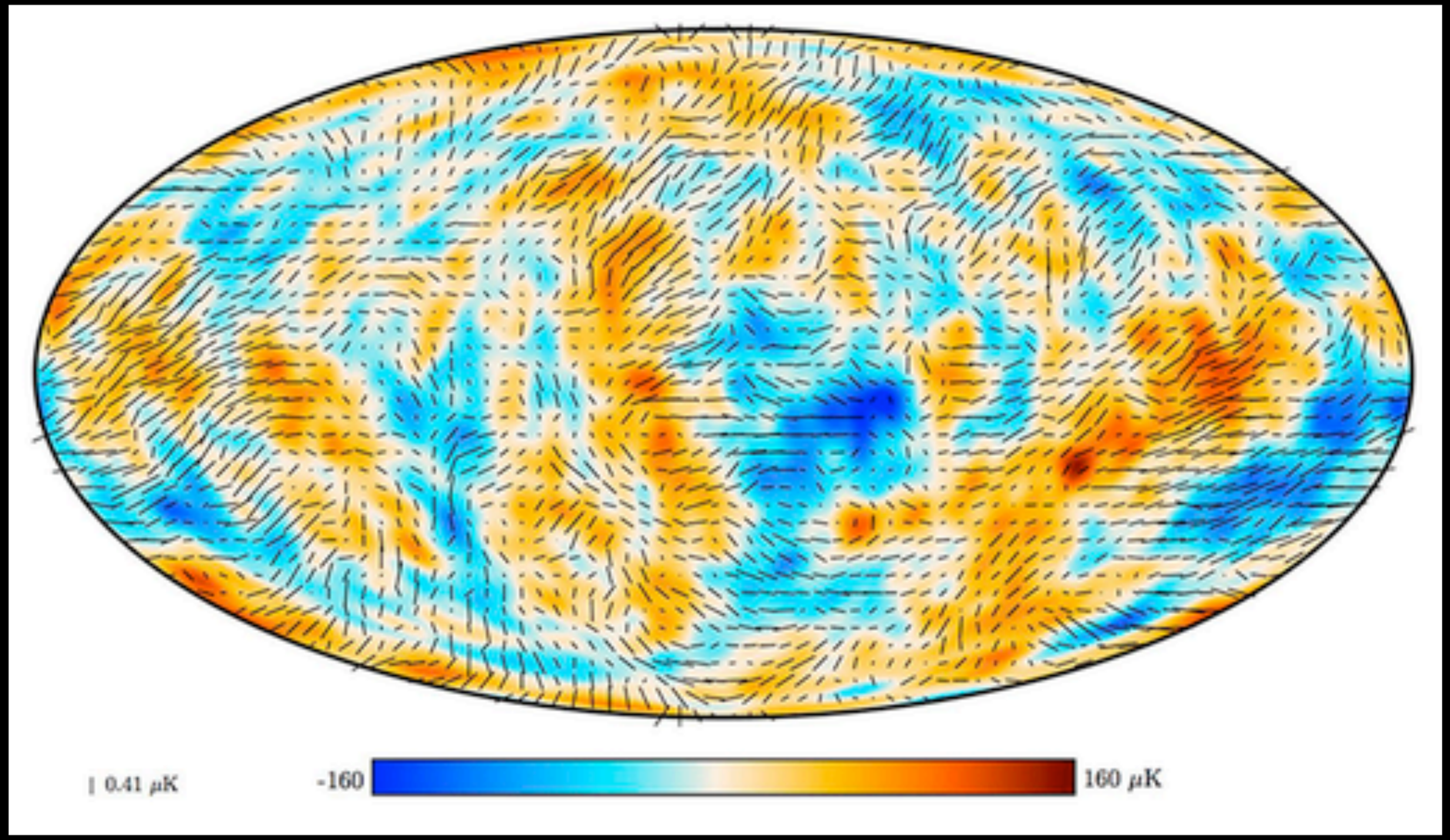




- Latest combined measurement from gravitational waves (GWTC-4):

$$H_0 = 76^{+13}_{-9.5}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$$

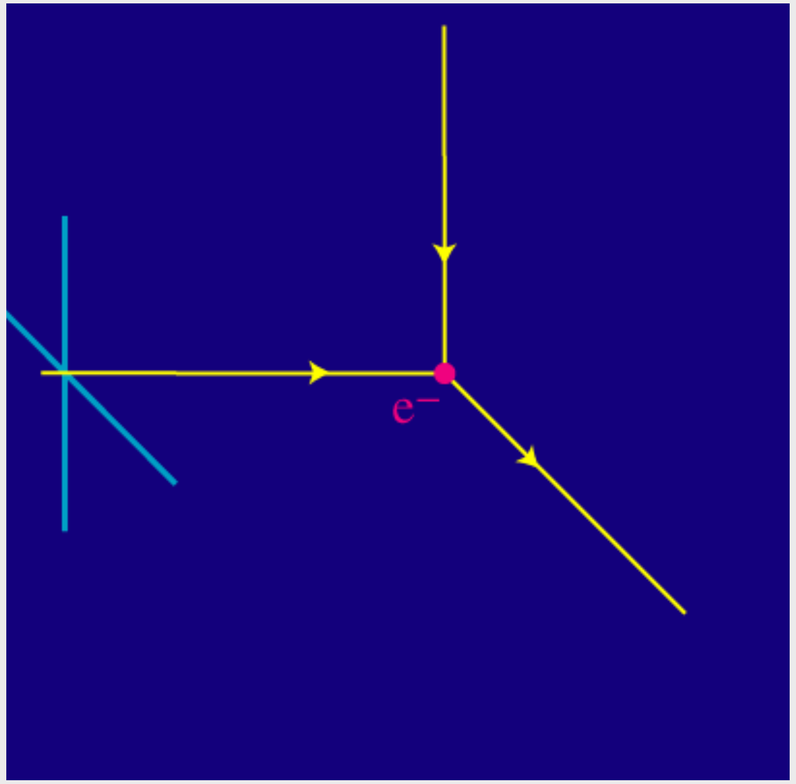
CMB POLARISATION



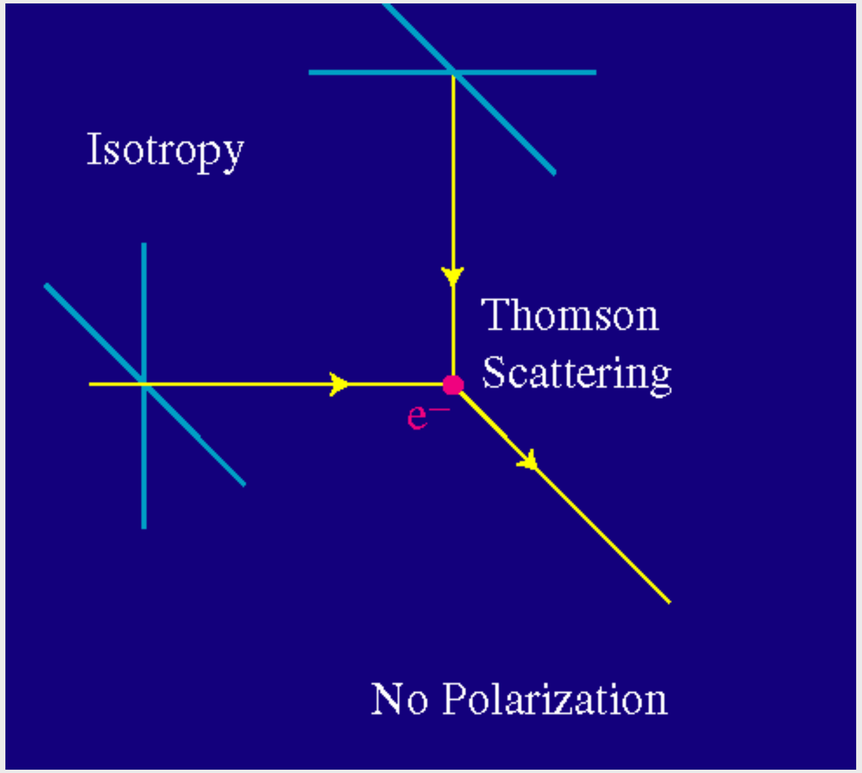
POLARISATION

Credit: Wayne Hu

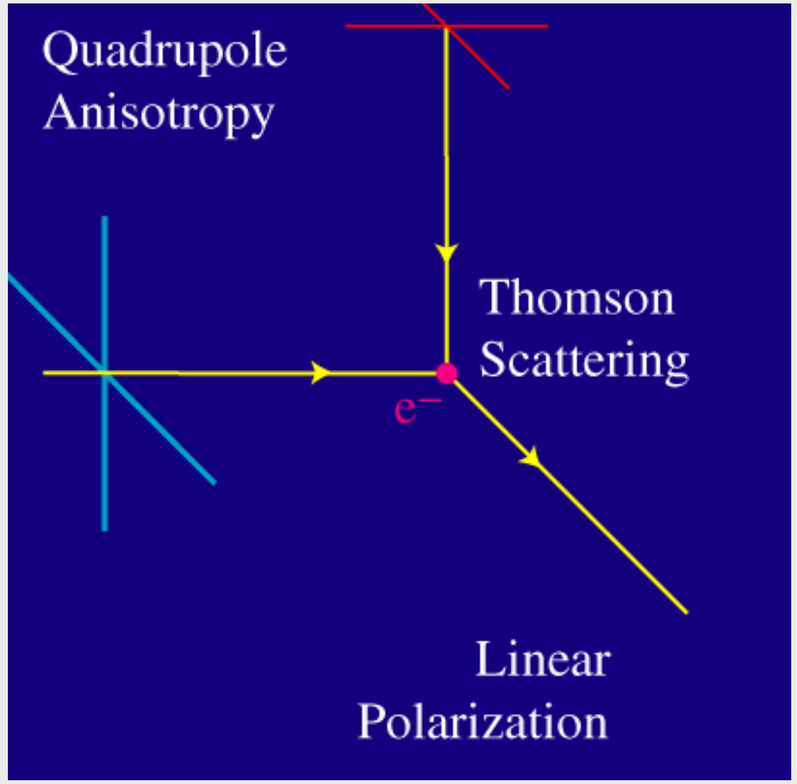
- Polarisation is exclusively **generated by scattering near recombination:**



Initially unpolarised light will be polarised by scattering.



The sum of all outgoing rays is unpolarised.

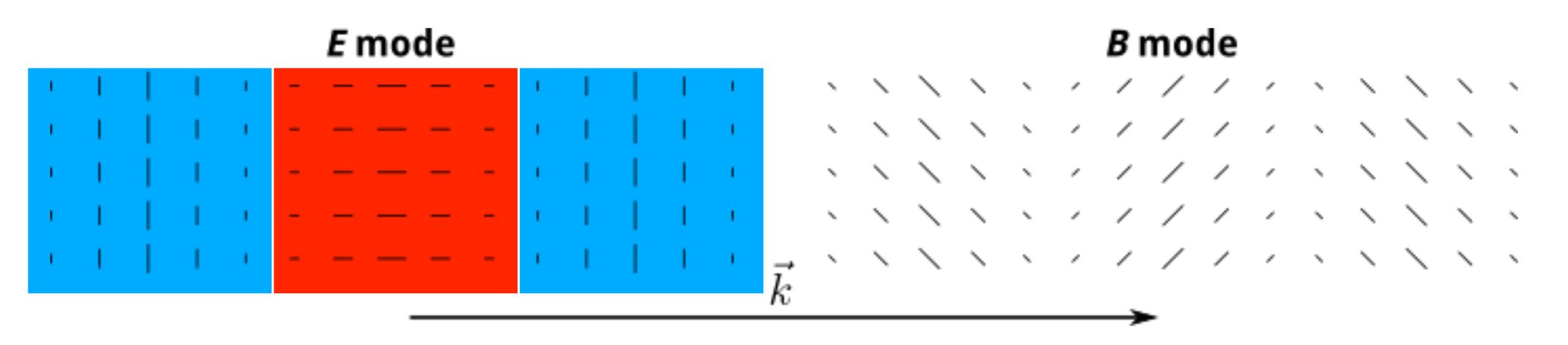


The outgoing radiation is linearly polarised.

The polarisation direction is aligned with the blue axis.

POLARISATION

- The observed polarisation pattern is the result of the **scattering of the temperature inhomogeneities** in the primordial plasma:
  - Consider a plane wave photon density perturbation propagating transverse to the line of sight



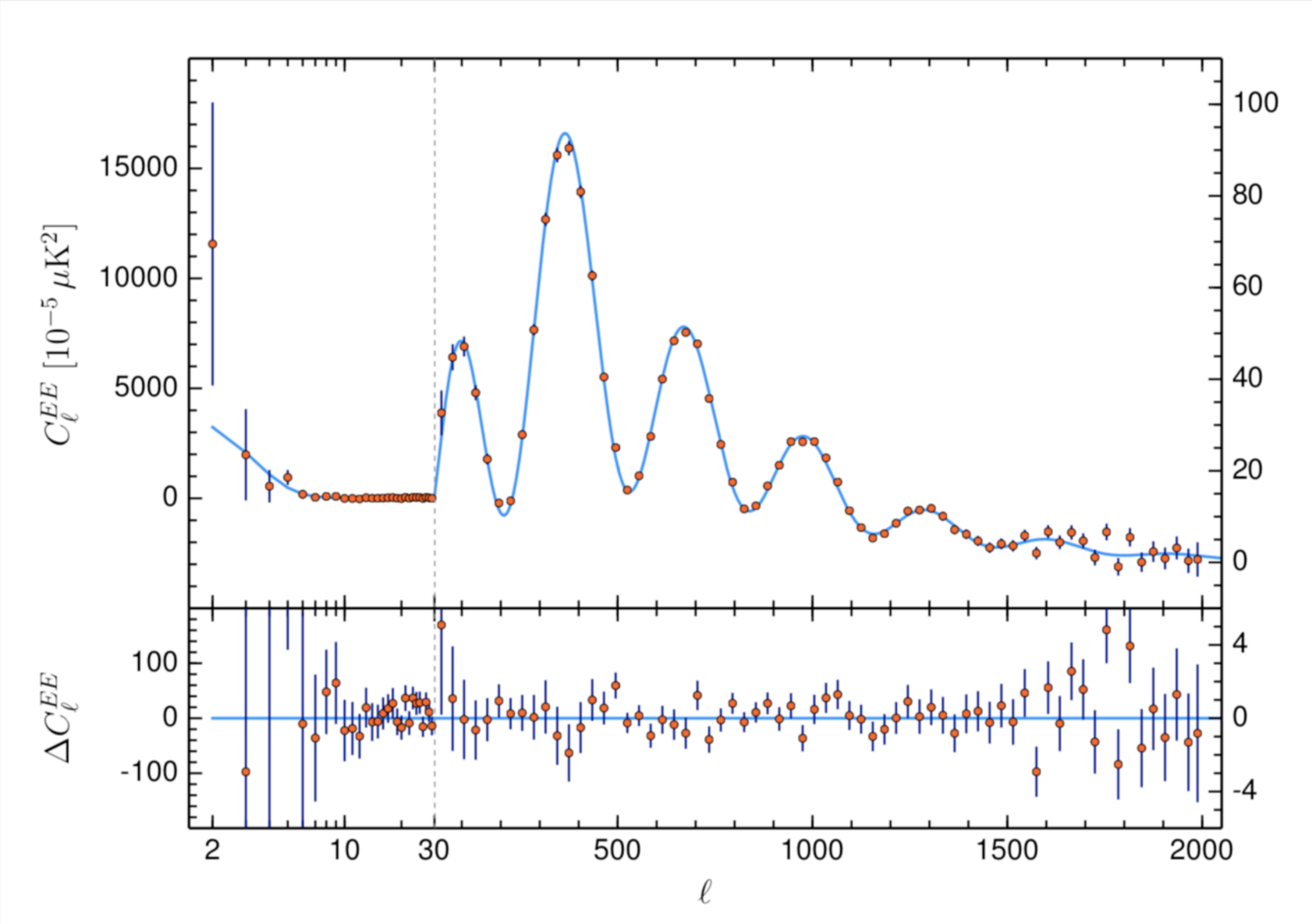
  - **E-mode** polarisation: induced by scalar perturbations, e.g. primordial density fluctuations
  - **B-mode** polarisation: generated by tensor perturbations, e.g. primordial gravitational waves

Smoking gun signal for inflation

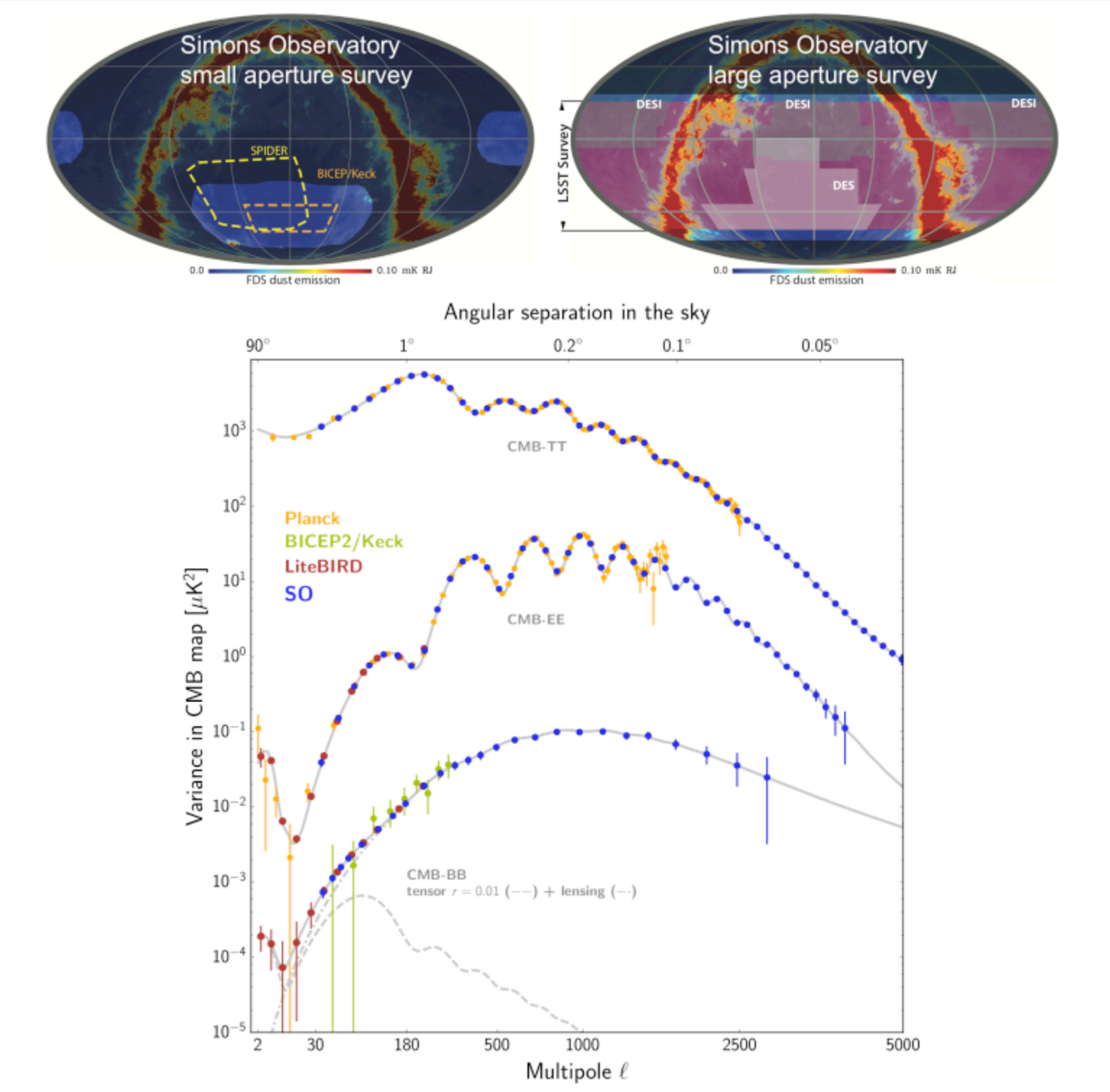
- The polarisation of electromagnetic radiation is defined by the **Stokes parameters** I, Q, U, V:
  - I measures the intensity; Q, U, V describe the polarisation state

EE POWER SPECTRUM

- Define polarisation power spectrum analogously to temperature power spectrum
- Note: Troughs in the polarisation power spectrum correspond to peak in the temperature power spectrum



THE BB POWER SPECTRUM



Measuring the BB power spectrum is a major science goal for the next-generation CMB missions